MILITARY SPECIFICATION SHEET

INSULATION SLEEVING, ELECTRICAL, HEAT SHRINKABLE, POLYVINYLIDENE FLUORIDE, SEMI-RIGID, CROSSLINKED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the sleeving described herein shall consist of this document and the latest issue of MIL-I-23053.

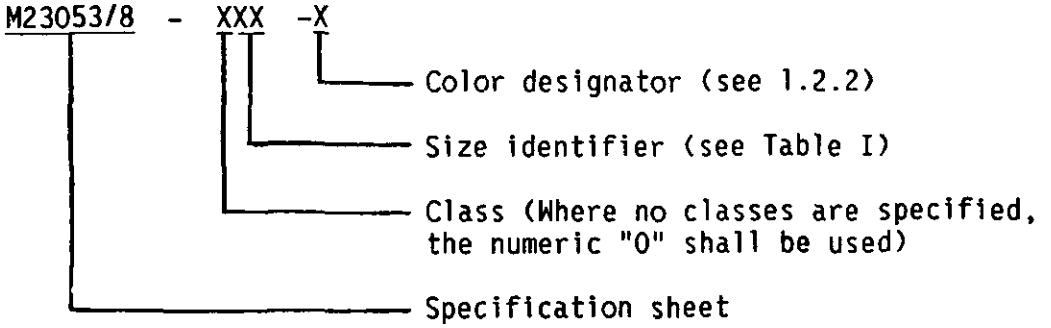
REQUIREMENTS:

Continuous operating temperature range: -55°C (-67°F) to +175°C (+347°F).

Color: The heat shrinkable sleeving shall be furnished transparent to translucent light tan (clear). Colors are available and shall conform to requirements of Class 1, MIL-STD-104.

Longitudinal change: ± 10 percent

Military part number: The Military part number shall consist of the basic number of this specification sheet and dash numbers as follows:



Example: 0.093 inch (2.36 mm) as supplied ID sleeving shall be identified as M23053/8-003-C.

NOTE: Entire specification sheet revised.

AMSC N/A FSC 5970

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TABLE I. Construction details, inches (mm). 1/

Military part number	As supplied ID minimum	After unrestricted shrinkage	
		ID maximum	Wall thickness 2/
M23053/8-001-C	.046 (1.17)	.023 (.58)	.010 ± .002 (.254 ± .051)
M23053/8-002-C	.063 (1.60)	.031 (.79)	.010 ± .002 (.254 ± .051)
M23053/8-003-C	.093 (2.36)	.046 (1.17)	.010 ± .002 (.254 ± .051)
M23053/8-004-C	.125 (3.18)	.062 (1.58)	.010 ± .002 (.254 ± .051)
M23053/8-005-C	.187 (4.75)	.093 (2.36)	.010 ± .002 (.254 ± .051)
M23053/8-006-C	.250 (6.35)	.125 (3.18)	.012 ± .003 (.305 ± .076)
M23053/8-007-C	.375 (9.53)	.187 (4.75)	.012 ± .003 (.305 ± .076)
M23053/8-008-C	.500 (12.70)	.250 (6.35)	.012 ± .003 (.305 ± .076)
M23053/8-009-C	.750 (19.05)	.375 (9.53)	.017 ± .003 (.432 ± .076)
M23053/8-010-C	1.000 (25.40)	.500 (12.7)	.019 ± .003 (.483 ± .076)
M23053/8-011-C	1.500 (38.10)	.750 (19.1)	.020 ± .003 (.508 ± .076)

1/ Diameter limits for object to be enclosed shall be as recommended in technical data.

2/ Wall thickness values are less when shrinkage is restricted.

Unrestricted shrinkage: Test method 4.6.5, 200° ± 2°C (392 ± 4°F) for 3 minutes.

TABLE II. Physical properties.

Characteristics	Requirement	Test procedure and conditions
As supplied:		
ID, minimum	Table I	4.6.3
Low temperature flexibility 3/	No cracking	4.6.7.1 -55° ± 1°C (-67° ± 2°F)
Heat shock	No cracks, flowing or dripping	4.6.8 300° ± 4°C (572° ± 7.2°F)
Secant modulus, psi (MPa), minimum	100,000 (690)	4.6.13.1 ASTM D 882, 2 percent strain
Restricted shrinkage	No cracks	4.6.6.1.1 175° ± 2°C (347° ± 4°F)
Voltage withstand	Pass	4.6.6.2
Clarity stability,	Pass	4.6.17 175° ± 2°C (347° ± 4°F) for 24 hours
Concentricity	70% min.	4.6.3.3

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TABLE II. Physical properties. - Continued

Characteristics	Requirement	Test procedure and conditions
After unrestricted shrinkage:		
ID, maximum	Table I	4.6.3
Wall thickness	Table I	4.6.3
Tensile strength, psi (MPa), minimum	5,000 (34.5)	4.6.14 ASTM D 638, 2 inches/minut
Ultimate elongation, percent, minimum	150	4.6.14 ASTM D 638, 2 inches/minut
Dielectric strength, volts/mil (Kv/mm), minimum	Sizes to .500 - 800 (31.5); Sizes over .500 - 600 (23.6)	4.6.2 ASTM D 2671
Volume resistivity, Ohm-cm, minimum	10^{13}	4.6.2 ASTM D 876
Specific gravity, max.	1.8	4.6.2 ASTM D 792
Water absorption, percent, maximum	0.5	4.6.2 ASTM D 570, 24 hrs at 23°C
Heat resistance, properties after:		4.6.9 250° ± 3°C (482° ± 5.4°F), 168 hours
Ultimate elongation, percent, minimum	50	
Fluid resistance, properties after:		4.6.11
Tensile strength, psi (MPa), minimum	5,000 (34.5)	
Dielectric strength, volts/mil (Kv/mm), minimum	500 (19.7)	
Flammability	Pass	4.6.15.3
Corrosion 1/	No corrosion	
Fungus resistance 2/	No growth	4.6.2 ASTM G 21

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TABLE II. Physical properties. - Continued

1/ MIL-I-23053/8 materials are non-corrosive at operating temperatures. Performance of this requirement is at the option of the requiring activity (see 6.2.1).

2/ MIL-I-23053/8 materials do not support fungus growth. Performance of this requirement is at the option of the acquiring activity (see 6.2.1).

3/ Low temperature flexibility test specimens and mandrels: Specimens of size 3/16 and smaller shall be tested as supplied without shrinking on conductors. Specimens of larger size shall be 1/4 inch wide strip cut from the tubing. Mandrels shall be 10 times the specimen O.D. ± 10% for tubular specimens or 10 times the specimen thickness ± 10% for strip specimens.

Shelf life conditions: Supplier shall certify to storage at 40°C (104°F) for four years. Conformance to 3.5. See 3.5.2 for shelf life extension.

Intended use: Heat shrinkable semi-rigid polyvinylidene fluoride sleeving is intended for wire or termination strain relief.

Custodians:
Army - ER
Navy - AS
Air Force - 85

Review activities:
Navy - EC
Air Force - 99
DLA - GS

Preparing activity:
Navy - AS

(Project 5970-0577-08)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-23053D/8B	INSULATION SLEEVING, ELECTRICAL, HEAT SHRINKABLE, GENERAL SPECIFICATION FOR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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